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Github link: https://github.com/idan-aizenberg/Final_project.git

WeatherSight- Long term weather forecast marketing platform

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Table of Contents

1. Abstract.....	1
2. Introduction.....	2
3. Literature Review.....	4
3.1 Background.....	4
3.2 Strategic Value of Long-Term Weather Forecasting.....	5
3.3 Strategic Advantages Across Key Industries.....	6
3.4 Existing Weather Forecasting Platforms.....	8
3.5 Data Presentation and Visualization Research.....	10
3.6 References.....	11
4. Expected Achievements.....	12
4.1 Technical Implementation.....	12
4.2 User Experience Design.....	13
4.3 Industry-Specific Applications.....	14
4.4 Business Model Implementation.....	15
4.5 Accessibility and Internationalization.....	16
5. System Architecture and Functional Requirements.....	17
5.1 Functional Requirements for Forecast Data Providers.....	17
5.2 Functional Requirements for the WeatherSight Platform.....	20
5.3 Subscription Tiers and Access Management.....	23
5.4 Administrative and Back-Office Functionality.....	25
6. System Architecture.....	26
6.1 Product Architecture.....	26
6.1.1 Architecture Diagram Description.....	26
7. Development Process.....	28
7.1 Methodology.....	28
7.2 Research and Conceptualization.....	28
7.3 Data Acquisition and Integration.....	28
7.4 Interface and Visualization Design.....	29
7.5 Backend Development.....	29
7.6 Alert and Notification System.....	29
7.7 Testing and Optimization.....	30
7.8 Finalization and Documentation.....	30
8. Testing Plan.....	31
8.1 Purpose.....	31
8.2 Objectives.....	31
8.3 Test Cases.....	32

1. Abstract

WeatherSight is a unified marketplace for long-term weather forecasting data, designed to bridge the operational disconnect between advanced climate model providers and the industries that depend on them. Today, while dozens of climtech firms are achieving breakthroughs in extended-range forecasting—leveraging innovations in ocean-atmosphere coupling, machine learning, and high-resolution modeling—these insights often remain trapped in fragmented ecosystems, underutilized due to poor distribution, inconsistent interfaces, or lack of sector relevance.

WeatherSight tackles three core challenges:

- **Data fragmentation** – climate data is dispersed across proprietary silos, making it difficult for users to compare, combine, or validate forecasts.
- **Lack of accessibility** – most model providers lack the infrastructure or resources to expose their data in business-friendly formats.
- **Actionability gap** – raw probabilistic forecasts are rarely translated into practical strategies tailored to the operational realities of industries like agriculture, logistics, energy, and insurance.

Our platform solves these problems through a multi-provider data integration layer, sector-specific visual analytics, and customizable decision-support tools. Users can compare forecasts, blend models, and receive alerts or recommendations based on thresholds relevant to their operations (e.g., planting dates, shipment windows, grid load planning). Forecast providers gain access to new revenue streams, real-world performance feedback, and a growing user base—without needing to build software distribution channels themselves.

By centralizing long-term weather intelligence in one robust, user-friendly platform, WeatherSight empowers industries to make climate-informed decisions, enabling both strategic foresight and day-to-day risk mitigation.

2. Introduction

In an era of increasing climate volatility, businesses across industries—from agriculture and energy to transportation and disaster preparedness—are desperate for reliable long-term weather insights. The problem? While cutting-edge climtech companies are making strides in weeks- and months-ahead forecasting, their breakthroughs often remain siloed, underutilized, or inaccessible to the very industries that need them most.

The Challenge: Great Data, No Distribution

Dozens of specialized firms worldwide are racing to develop the most accurate long-term weather models, each with unique strengths—whether in ocean-atmosphere coupling, machine learning enhancements, or regional precision. Yet these innovators face a critical bottleneck: very few have the resources or expertise to build their own customer-facing platforms. Even if they did, the market risks becoming oversaturated with niche solutions, leaving end-users confused and reluctant to adopt any single tool.

Our Solution: A Unified Marketplace for Forecast Intelligence

We're building the dominant platform for long-term weather data—a centralized hub where businesses can access, compare, and apply forecasts from multiple leading providers. Instead of forcing climtech companies to become sales and software experts, we handle distribution, visualization, and industry-specific insights—letting them focus on what they do best: improving their models.

For users, this means:

- Choice – Select forecasts from top models or blend insights from multiple sources.
- Clarity – Intuitive dashboards tailored to sector-specific needs
- Confidence – Probabilistic trends translated into actionable strategies, not just raw data.

For forecast providers, this means:

- Revenue – Monetize data without building costly in-house platforms.
- Reach – Tap into a broad customer base that trusts our unified interface.
- Feedback – Gain real-world performance metrics to refine models.
- Why One Platform Must Win

By connecting advanced forecasting science with practical applications, this platform empowers industries to build climate resilience. We facilitate access to the best available predictions while helping users translate complex data into actionable strategies - all through a single, unified interface.

3. Literature Review

3.1 Background

The capability to access accurate and reliable long-term weather forecasts is essential for diverse stakeholders reliant on environmental information for critical decision-making processes. Unlike short-term forecasts focused on imminent conditions, long-term predictions span months or even a year, encompassing probabilistic outcomes that significantly influence strategic planning and risk management¹. The accessibility and interpretability of such forecasts remain a considerable challenge, as most current weather services prioritize short-term accuracy or deliver complex, technical data unsuitable for non-expert users. Addressing this gap through intuitive, user-specific platforms promises enhanced societal decision-making, benefiting individual users and complex organizational structures alike².

3.2 Strategic Value of Long-Term Weather Forecasting

Long-term weather forecasts are vital across various sectors, including agriculture, energy, logistics, construction, and insurance. In agriculture, they assist farmers in optimizing planting schedules, irrigation planning, and resource allocation, directly influencing crop yields and economic sustainability. Energy producers utilize extended forecasts to anticipate demand fluctuations and manage renewable energy systems, thereby enhancing operational efficiency and grid reliability. Similarly, accurate long-range forecasting informs logistical planning, infrastructure project timelines, and disaster mitigation strategies, reducing vulnerability to adverse weather impacts. The insurance industry also relies on these forecasts to assess risk and set premiums accurately, particularly for policies related to agriculture, property, and disaster coverage. Overall, long-term weather forecasting serves as a critical component in strategic planning and risk management across multiple industries, contributing to economic stability and resilience against climate variability.³

¹ Sivakumar & Hansen, "Climate predictions and agriculture"

² Tall, coulibaly, Diop, "Do climate services make a difference"

³ Li, Liu, Chen, Wu, "Enhanced Agriculture Insurance with Climate Forecast"

3.3 Strategic Advantages of Long-Term Weather Forecasting Across Key Industries

Long-term weather forecasting plays a pivotal role across various sectors, enabling strategic planning and effective risk mitigation.

Agriculture

Farmers utilize long-term forecasts to optimize planting schedules, irrigation practices, and resource allocation, which enhances productivity and reduces vulnerability to adverse climatic conditions.⁴

Urban Planning

Urban planners use long-range weather forecasts to design resilient infrastructure and adapt city layouts, thereby ensuring public safety and enhancing sustainability amid climate variability.⁵

Real Estate

Real estate professionals leverage long-term weather forecasts for informed investment decisions, property valuations, and development strategies, safeguarding properties against future climatic risks.⁶

Insurance

Insurers apply extended weather forecasts to enhance actuarial accuracy, refine pricing strategies, and effectively manage exposure to climate-induced financial risks.⁷

Logistics and Supply Chain

Supply chain managers depend on long-term weather forecasting to anticipate disruptions, optimize transportation routes, and sustain operational efficiency, thus ensuring timely delivery and customer satisfaction.⁸

Energy Sector

Energy providers, especially those managing renewable energy resources, utilize long-term forecasts to predict energy production patterns, balance grid demands, and schedule maintenance, thereby stabilizing energy supplies and optimizing performance.⁹

⁴ Burlig, Jina, Kelley, Lane, Sahai, "Better Weather Forecasts Can Help Farmers Adapt to Climate Change"

⁵ Zeng, "Climate Resilience with AI-Powered Weather Forecast"

⁶ Zeng, "Climate Resilience with AI-Powered Weather Forecast"

⁷ Li, Liu, Chen, Wu, "Enhanced Agriculture Insurance with Climate Forecast"

⁸ Acumen Research, "Weather Forecasting Services Market Size Growing at 9.4% CAGR, Set to Reach USD 6.1 Billion By 2032"

⁹ Climavision, "How Weather Impacts Energy Trading and Tools to Help Mitigate Risk"

3.4 Existing Weather Forecasting Platforms

Short-Term Platforms:

AccuWeather: Offers forecasts up to 45 days ahead with a strong focus on user-friendly design, intuitive dashboards, and visualized weather trends for consumers and businesses.

<https://www.accuweather.com>

Weather.com: The Weather Channel's platform integrates interactive visuals, radar imagery, and hyper-local forecasting tailored to users' locations.

<https://www.weather.com>

OpenWeatherMap: Provides detailed weather data through an API, supporting integration into applications and platforms for forecasts up to 5 days.

<https://openweathermap.org>

Open-Meteo: An open-source platform focused on free, high-resolution, customizable weather forecasts for up to 35 days, designed for developers and analysts.

<https://open-meteo.com>

Windy.com: Specializes in dynamic, high-resolution interactive weather maps, popular among professionals such as pilots, sailors, and outdoor enthusiasts.

<https://www.windy.com>

Long-Term Platforms:

WeatherPlanner: Provides forecasts up to one year in advance based on historical patterns and statistical models, targeting event planning and agriculture.

<https://www.weatherplanner.com>

Weather 20/20: Uses the Lezak Recurring Cycle (LRC) to identify atmospheric patterns that repeat seasonally, enabling long-term strategic forecasting.

<https://www.weather2020.com>

WeatherTrends360: Employs statistical analysis of decades of historical weather data to generate predictive models used widely in retail, finance, and agriculture.

<https://www.weathertrends360.com>

ECMWF: The European Centre for Medium-Range Weather Forecasts provides scientific seasonal outlooks using ensemble models to predict anomalies up to 13 months ahead.

<https://www.ecmwf.int>

Farmers' Almanac: Offers long-term forecasts based on a proprietary methodology involving astronomical cycles and tide patterns, often used for seasonal farming and outdoor events.

<https://www.farmersalmanac.com>

3.5 Data Presentation and Visualization Research

Effective visualization is pivotal in enhancing user comprehension and facilitating decision-making in weather forecasting. Key principles include:

Clarity: Presenting data in an unambiguous manner to prevent misinterpretation.

Contextual Relevance: Make sure the visuals match what the user needs and the context they're in.

Effective Communication of Uncertainty: Make it clear how reliable the forecasts are and how much they might vary.

Interactive visualization platforms are particularly beneficial for non-specialist audiences. They allow users to engage dynamically with data, tailoring the information to their specific requirements. This interactivity enhances understanding and supports informed decision-making.

Modern weather forecasting platforms increasingly rely on advanced visualization systems to present large volumes of complex data in an accessible and actionable format. One such implementation is a web-based platform that integrates observational data, numerical weather predictions, and satellite imagery into a unified visualization and analytical environment. This platform was designed to support short-term forecasting and model validation by enabling direct comparisons between predicted and observed atmospheric parameters, such as temperature, humidity, wind, and precipitation.

The system leverages Spatial OLAP (SOLAP) for multidimensional data analysis, allowing users to perform spatial-temporal queries and view aggregated results across stations and regions. The visualization interface combines interactive maps, trend charts, and statistical panels to communicate forecast accuracy. Components include a

query form, real-time map layer, temperature graphs, and tables displaying key metrics like RMSE, bias, and correlation coefficients.

The platform architecture incorporates PostGIS-enabled PostgreSQL databases, Google Maps for spatial rendering, and GRASS GIS for satellite image processing. Data ingestion and transformation are handled using Python and JavaScript scripts, ensuring automation and scalability.

Future enhancements include expanding the platform to cover more meteorological variables, integrating AI-based nowcasting, and connecting with cloud services for distributed access and processing. This example illustrates the critical role of interactive data visualization in improving model transparency, operational forecasting, and decision support across sectors like agriculture, urban planning, and disaster risk reduction.¹⁰

3.7 References

- [Climate Prediction and Agriculture: Advances and Challenges | SpringerLink](#)
- [Do climate services make a difference? A review of evaluation methodologies and practices to assess the value of climate information services for farmers: Implications for Africa - ScienceDirect](#)
- [Enhanced Agriculture Insurance with Climate Forecast](#)
- [Better Weather Forecasts Can Help Farmers Adapt to Climate Change | EPIC](#)
- [Climate Resilience with AI-Powered Weather Forecast](#)
- [Weather Forecasting Services Market Size Growing at 9.4% CAGR, Set to Reach USD 6.1 Billion By 2032](#)
- [Weather's Impact on Energy Trading & Risk Mitigation](#)
- [\(PDF\) Weather Data Visualization and Analytical Platform](#)

¹⁰ Astsatryan, Gyulgyulyan "Weather Data Visualization and Analytical Platform"

4. Expected Achievements

4.1 Technical Implementation- The WeatherSight platform will achieve a robust technical foundation through the development of a scalable data integration framework capable of processing and harmonizing long-term weather forecast data from multiple meteorological sources. This system will maintain data integrity while minimizing latency, ensuring users receive timely and accurate information. The platform will implement a user-adaptive visualization engine that dynamically adjusts complexity and presentation based on user classification and industry context, making complex meteorological data accessible to users of varying technical backgrounds.

Cross-platform accessibility will be achieved through responsive design principles, enabling consistent user experience across desktop, mobile, and tablet devices without compromising functionality or visual integrity.

4.2 User Experience Design- The platform will deliver an intuitive and streamlined user experience through carefully designed interfaces tailored to different user types. We will develop customizable dashboard systems allowing users to configure their view of forecast data according to their specific needs and preferences. The platform will implement an innovative uncertainty visualization framework that presents probabilistic forecast information in ways that are intuitive to non-technical users while preserving scientific accuracy. User engagement will be optimized through iterative design based on usability testing with representatives from each target user segment, ensuring the platform meets the practical needs of diverse users from individuals planning personal events to corporations managing complex operations.

4.3 Business Model Implementation- The project will establish a viable multi-tiered subscription model accommodating various user types from individuals to large corporations, with appropriate feature access and pricing for each tier. This model will balance accessibility for individual users with premium capabilities for enterprise clients, creating a sustainable revenue structure. The platform will integrate secure payment processing and subscription management systems with automated billing and renewal capabilities.

4.4 Accessibility and Internationalization- The platform will be developed with accessibility standards to ensure inclusive use by individuals with disabilities. Features will include keyboard navigation, screen reader compatibility, color-blind friendly palettes, and adjustable font sizes. To support a global user base, WeatherSight will be internationalized, enabling easy localization into multiple languages and regional formats (e.g., Celsius vs. Fahrenheit, date/time formats). This will increase adoption across diverse markets, making the platform relevant and usable for a broad spectrum of individual and institutional clients worldwide.

5. System Architecture and Functional Requirements

5.1 Functional Requirements for Forecast Data Providers

To ensure robust performance, interoperability, and forecast reliability, WeatherSight must interface seamlessly with external data providers offering long-range meteorological models. The following functional requirements define the expected capabilities from these forecast data sources:

A. Forecast Range

1. The data provider shall support forecasts extending up to 365 days into the future.
2. Forecasts must accommodate both deterministic values and probabilistic distributions.
3. Forecasts must support all known locations possible.

B. Forecast Variables and Event Probabilities

The platform must ingest a comprehensive array of meteorological variables, including:

Standard Forecast Parameters:

1. Minimum, maximum, and average temperature.
2. Precipitation volume and probability.
3. Wind speed and direction.
4. Humidity and dew point.
5. Solar radiation and UV index.
6. Air pressure.

Probabilistic Forecast Data:

1. Probability distributions (e.g., chance of rainfall, frost, drought)
2. Confidence intervals and statistical uncertainty indicators

Extreme Weather Event Metrics:

Forecasts shall also include probabilities and severity classifications for the following event types:

1. Heatwaves / Cold spells
2. Sandstorms, haze, or air quality degradation
3. Thunderstorms and lightning strikes
4. Heavy precipitation, floods, and flash floods
5. Snowstorms and freezing conditions
6. Tropical storms, hurricanes, and tornadoes

Each extreme event entry must be accompanied by:

1. Event probability (0–1 or percentage scale)
2. Severity level (low, medium, high; or numeric rating)
3. Estimated event duration and onset time window

C. Reliability and Availability

1. The provider must guarantee that the system remains available at least 99.9 percent of the time.
2. Forecast data must be updated Weekly or more frequently where possible.

5.2 Functional Requirements for the WeatherSight Platform

The WeatherSight platform serves as a middleware layer between climate data providers and end-users in various industries. Its primary function is to transform scientific forecasts into accessible, industry-relevant insights.

A. User Account Management

1. The platform shall support secure user registration and authentication using email and password.
2. Users will configure profile preferences such as default location, industries of interest, and desired forecast resolution.
3. The system shall differentiate between free-tier and premium-tier accounts, managing access and feature availability accordingly.

B. Query Interface and Forecast Input

Users will interact with a structured, intuitive query interface enabling the following inputs:

1. Location selection via text input or interactive map
2. Date range selection (ranging from 7 to 365 days)
3. Resolution: Daily, weekly average, or monthly summary
4. Forecast type: Standard metrics, probabilistic distributions, or extreme weather events
5. Output format: Tabular view, interactive chart, or downloadable file (CSV/Excel)

C. Visualization and Forecast Presentation

1. Forecast results shall be displayed via interactive dashboards featuring:
 - 1.1 Line graphs, heat maps, and probability bars
 - 1.2 Severity icons and color-coded risk levels
2. Users will be able to toggle between raw data, statistical summaries, and custom visualizations tailored to their needs.

D. Query Persistence and User History

1. Users shall be able to save queries for future reuse.
2. A query history panel shall allow users to review past requests, including results and timestamp.

E. Alerts and Notifications (Paid Feature)

1. Non-free tier users may define custom alert thresholds (e.g., >70% chance of frost).
2. Alerts will be delivered via email, SMS, or push notifications.
3. Users will configure notification frequency (e.g., daily summaries, immediate alerts).

5.3 Subscription Tiers and Access Management

The platform's feature availability and query limits are governed by a tiered access model:

Basic (Free) Tier

1. Up to 3 forecast queries per day
2. Forecast horizon limited to 14 days
3. Access to basic weather metrics (e.g., temperature, precipitation)
4. No access to probabilistic models or extreme weather data
5. No export or notification features

Standard Tier

1. Up to 10 queries per day
2. Forecast horizon up to 30 days
3. Access to additional metrics (e.g., humidity, wind, solar radiation)
4. Limited access to probabilistic trends (no multi-model comparison)
5. No alert system included

Professional Tier

1. Up to 100 queries per day
2. Forecast horizon up to 180 days
3. Full access to probabilistic forecasts and visual confidence intervals
4. Access to moderate-risk extreme weather predictions
5. Export reports in CSV, PDF, and Excel formats
6. Custom alert configuration and delivery via email
7. Access to one industry-specific dashboard

Enterprise Tier

1. Unlimited queries per day
2. Forecast horizon up to 365 days
3. Full access to advanced probabilistic models and multi-model blending
4. Real-time alerts for high-risk extreme weather events (SMS, Email, Push)
5. Access to all industry-specific dashboards (agriculture, logistics, etc.)
6. Priority processing (QoS) and API integration for enterprise systems
7. Dedicated support and account management

5.4 Administrative and Back-Office Functionality

A. Admin Console

Platform administrators will manage:

1. Subscription and billing tiers
2. Logging and diagnostics

B. Performance and Monitoring Tools

1. Real-time usage monitoring dashboards
2. System health reporting and scalability testing tools

6. System Architecture

6.1 Product Architecture

The architecture of WeatherSight will be designed to provide an end-to-end solution for long-range weather forecasting access and analysis. The system will be composed of four main components: data ingestion and integration, backend processing, frontend visualization, and notification services. This modular structure will support scalability, maintainability, and integration with external weather forecast providers.

6.1.1 Architecture Diagram Description

Data Ingestion Layer – Will periodically collect data from multiple weather forecast providers using standard data exchange methods. The data will be processed and stored in a structured, normalized format for efficient access.

Backend Services – Will manage user authentication, handle query processing, support forecast model integration, and generate alerts. These services will operate through secure APIs and ensure reliable system logic and workflow management.

Database Layer – Will store structured forecast data, user profiles, query histories, and alert configurations. It will support geospatial capabilities to enhance location-based querying and analysis.

Frontend Interface – Will be implemented as a responsive web application that delivers an interactive user experience. It will include visual dashboards, data input forms, and customizable views tailored to user preferences and subscription levels.

Notification Layer – Will operate asynchronously to monitor forecast thresholds and deliver alerts through various communication channels such as email and SMS. This layer will ensure timely dissemination of critical information to users based on their configured settings.

7. Development Process

7.1 Methodology

WeatherSight will be developed using an Agile methodology to support iterative progress and modular development. The project will be divided into two academic semesters: the first semester will focus on system design, architecture, and research activities, while the second will cover the core development, system integration, and optimization.

7.2 Research and Conceptualization

The development process will begin with the identification of key requirements for long-term weather forecast accessibility. Our team will analyze industry needs and examine best practices in data visualization and interface design. Based on this analysis, system requirements and architectural plans will be defined.

7.3 Data Acquisition and Integration

A specific and reliable weather forecast provider has been selected to supply long-range meteorological data. The team will implement data ingestion routines to regularly collect, standardize, and store forecast data. These routines will be designed to ensure accuracy, consistency, and update frequency in accordance with system requirements.

7.4 Interface and Visualization Design

The interface will be designed with clarity, responsiveness, and accessibility in mind. Design prototypes will guide implementation of the main screens, ensuring that forecast results and probabilistic outputs are presented in a clear and usable format across desktop and mobile platforms.

7.5 Backend Development

The backend will be responsible for secure user authentication, query processing, data access, alert evaluation, and subscription management. Backend logic will ensure efficient data handling and system stability.

7.6 Alert and Notification System

The alert system will monitor forecast data in real time and match it against user-defined thresholds. When a threshold is met, alert messages will be dispatched via email or

SMS. This system will operate independently to ensure minimal latency in the alert workflow.

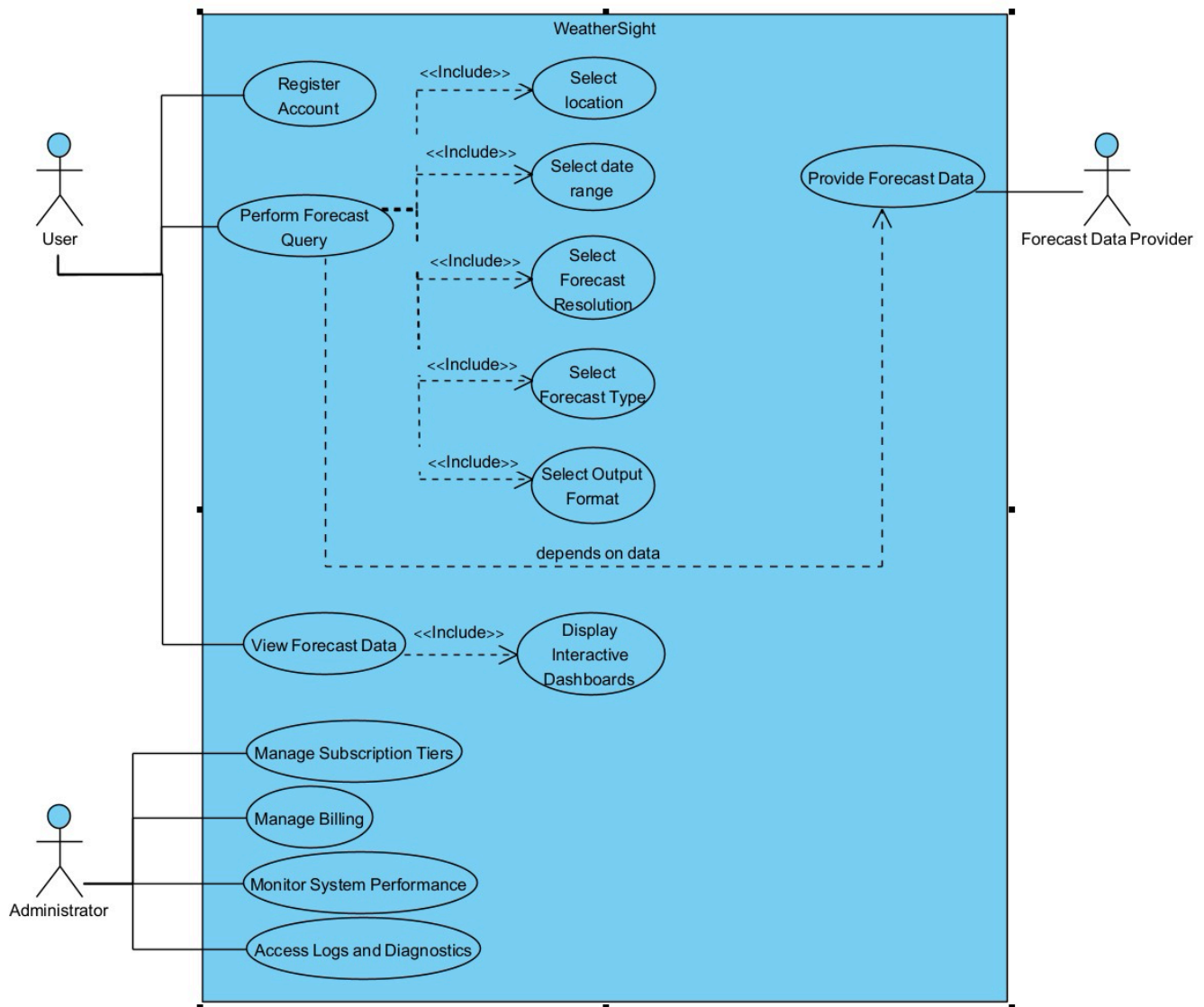
7.7 Testing and Optimization

The team will execute unit tests, integration tests, and performance benchmarks during development. Emphasis will be placed on system stability, correctness, and responsiveness under simulated load conditions. Optimization efforts will focus on improving query speed, data handling efficiency, and scalability.

7.8 Finalization and Documentation

In the final stage of development, the system will undergo final revisions, error correction, and deployment preparation. Full documentation of components, system architecture, and operating procedures will be compiled to ensure maintainability and ease of onboarding for future development efforts.

7.9 Use Case diagram



User Interface:

Home Page- The user lands on the Home Page, sees the value proposition and clicks “Get Started.”

Dashboard- After logging in, they arrive at their Dashboard, where recent searches, and a “New Search” button await.

Search Page- Clicking “New Search” opens the filter panel: they pick location, date range, and comfort/risk presets.

Results Page- Submitting the search brings up interactive charts, heatmaps, and weekly forecast cards—then they can save the search or set an alert.

Home Page:

WeatherSight

ProductSolutionsPricingResourcesGet StartedLogin

Predict the Future, Plan with Confidence

WeatherSight provides advanced weather forecasting tools for businesses and individuals, enabling proactive planning and informed decision-making.

🔍 Enter your location

Search

Empowering Proactive Planning

WeatherWise provides the tools and insights you need to stay ahead of the weather, minimize risks, and maximize opportunities.



Hyperlocal Forecasting
Get precise weather forecasts for specific locations, down to the neighborhood level.



Long-Range Planning
Plan weeks, months, or even years ahead with our long-term forecasting capabilities.



Advanced Analytics
Analyze historical weather data, identify trends, and optimize your operations.

Analyze historical weather data, identify trends, and optimize your operations.

Ready to Take Control of the Weather?

Start your free trial today and discover how WeatherWise can transform your planning and decision-making.

Get Started

Comprehensive Weather Intelligence



Global Coverage
Access weather forecasts for any location worldwide, with detailed information and insights.



Real-Time Updates
Stay informed with up-to-the-minute weather updates, ensuring you have the latest information.



Customizable Alerts
Set personalized alerts for specific weather conditions, ensuring you never miss important changes.



Historical Data
Analyze historical weather patterns to identify trends, understand risks, and optimize your strategies.



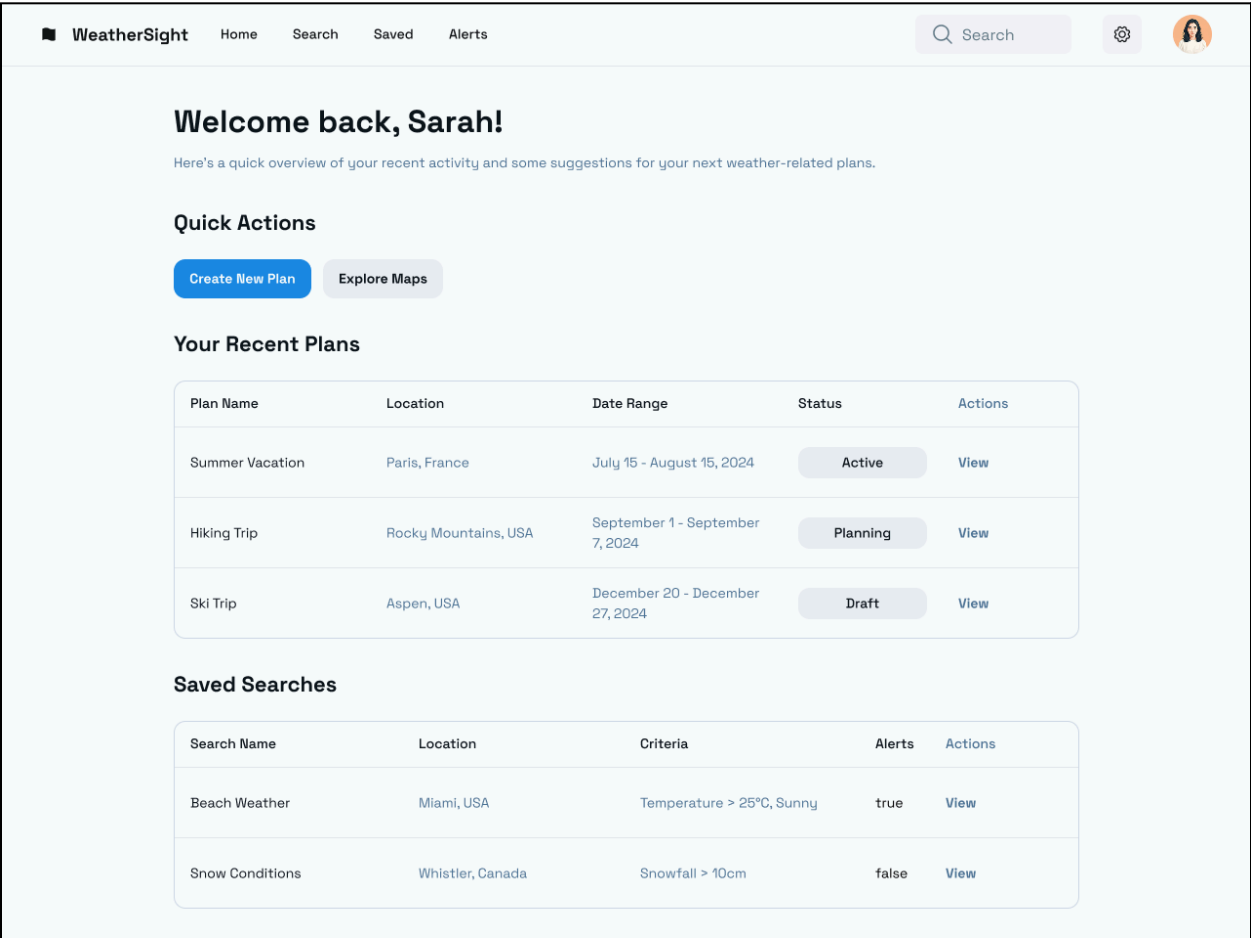
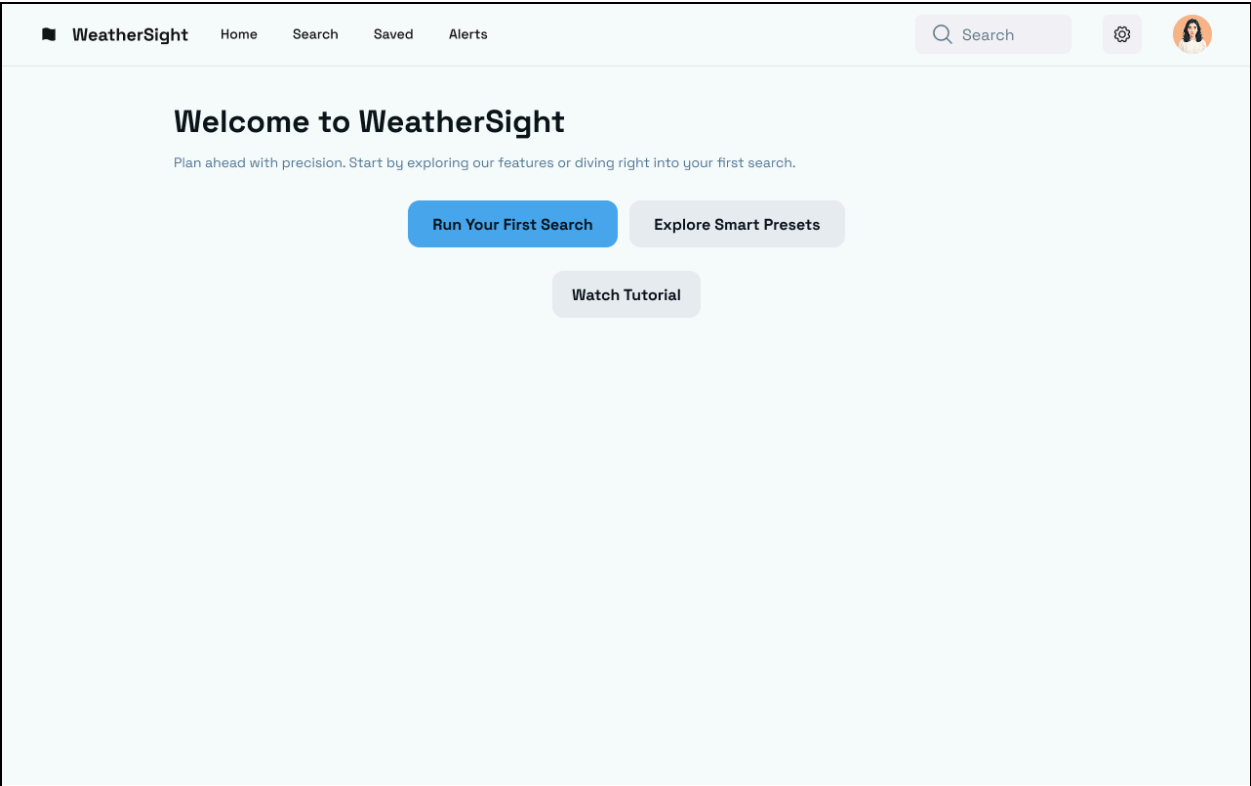
API Integration
Integrate WeatherWise data seamlessly into your existing systems and workflows.

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Dashboard Page (for first time user and registered user):



Search Query Pages:

Search by specific conditions to find times when they're met:

Search Query Pages:

Search by specific conditions to find times when they're met:

Search Query Pages:

Search by specific conditions to find times when they're met:

Search expected forecast in specific location:



Search distributions and probabilities:

WeatherSight

Home

Search

Saved

Alerts

Search

Expected

Distribution

Time Finder

Central Parameter Builder

+ Add another parameter

Histogram

Box-and-Whisker

CDF

Percentile

Show chance ≥ 30 °C exceeds

Statistic	Value
Min	18°C
Max	35°C
Mean	26°C
Median	25°C
P25	22°C
P75	28°C

Save Search

Search

General Search Page:

WeatherSight

Home

Search

Saved

Alerts

Search



Weather Search

Basic Requirements

Location

Enter city or zip code

Start Date

Select start date

End Date

Select end date

Purpose

Weather Parameters

Temperature Range (°C)

Rain Chance

20%

Precipitation

5mm

Wind Speed (km/h)

Humidity

60%

UV Index

7

Cloud Cover

30%

Comfort & Risk

Comfort Index

70

Severe Weather Risk

20

Risk Tolerance for Storms

Risk Tolerance for Heatwaves

Risk Tolerance for Cold Snaps

Risk Tolerance for High Winds

Risk Tolerance for Heavy Rain

Smart Presets

Sunny Day

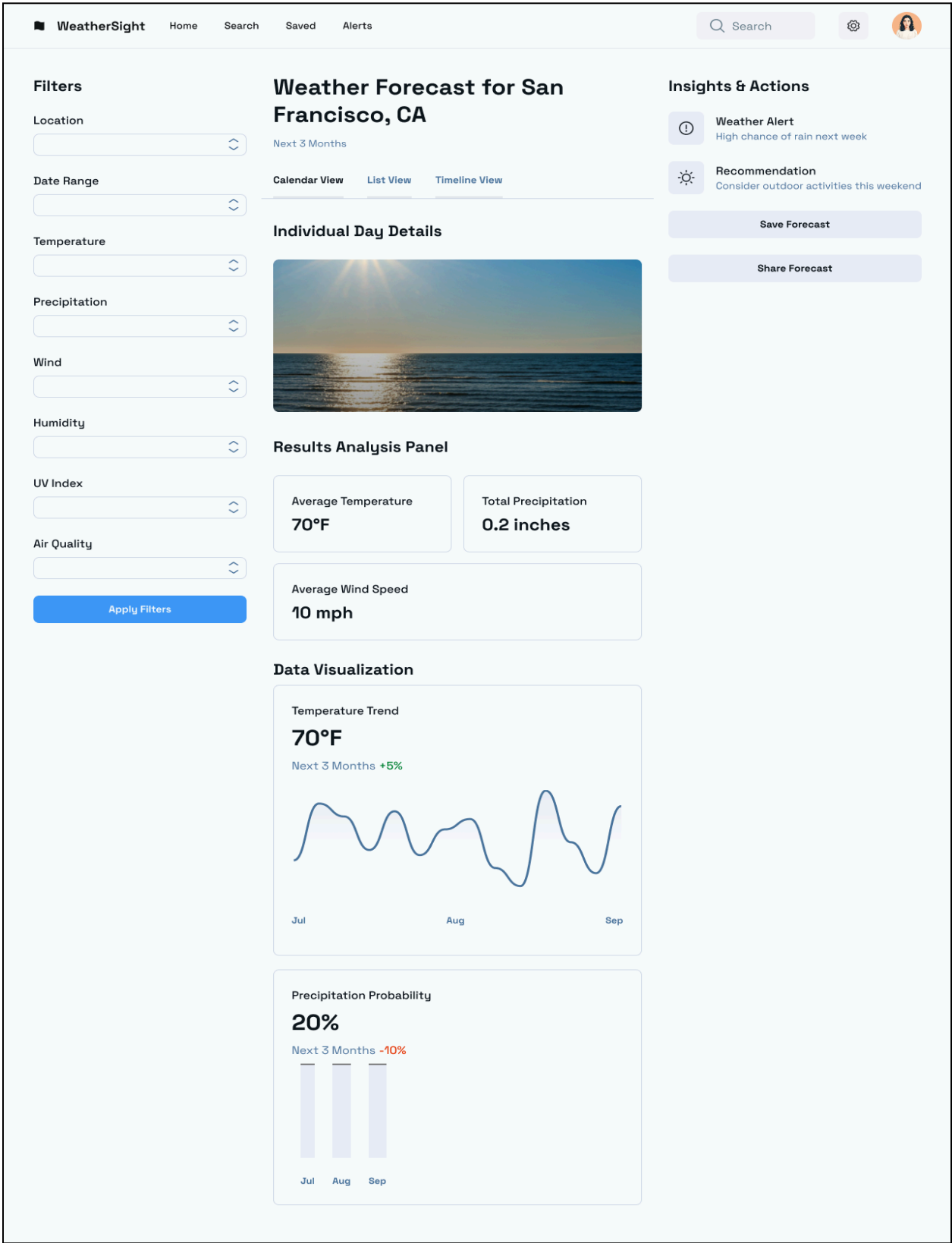
Mild Weather

Low Risk

Save Search

Search

Results Page:



Temperature Trend

70°F

Next 3 Months +5%



JulAugSep

Precipitation Probability

20%

Next 3 Months -10%



JulAugSep

8. Testing Plan

8.1 Purpose

The purpose of the testing plan is to ensure the WeatherSight platform meets its functional, performance, and usability requirements. Through systematic testing, the platform will be validated against real-world usage scenarios to verify it functions correctly, is user-friendly, and can handle long-term weather forecast processing at scale. This plan is essential to identify and resolve issues before deployment to production environments.

8.2 Objectives

The primary objective of the testing phase is to verify that the WeatherSight platform performs reliably and meets its intended design goals. The testing will focus on ensuring that all features—from user authentication to query execution and visualization—operate as expected. It will confirm that different user tiers are handled correctly, data is visualized clearly, alerts are delivered on time, and the platform remains stable under various usage conditions. Additional emphasis will be placed on user experience, ensuring the system is intuitive, responsive, and accessible across different devices.

8.3 Test Cases

Test Case Number	Test Scenario	Test Steps / Action	Expected Result
1	User signs up successfully	1. Open sign-up form 2. Enter valid email and password 3. Submit form	Account is created, and user is redirected to dashboard
2	User logs in with valid credentials	1. Open login form 2. Enter registered email and correct password 3. Submit	User is authenticated and redirected to their dashboard
3	User fails login with wrong password	1. Open login form 2. Enter correct email and wrong password 3. Submit	System shows error message: 'Invalid credentials'

4	User resets password	1. Click 'Forgot Password' 2. Enter email 3. Reset link 4. Change pass	Password is updated and user can log in with new credentials
5	User edits profile preferences	1. Login 2. Go to Profile 3. Change settings 4. Save	Preferences are saved and reflected in future queries
6	Submit a valid 30-day weather query (Free tier)	1. Login 2. Open query 3. Select location 4. Set range 5. Submit	Query returns basic metrics for selected range
7	Attempt 4th query as Free user	1. Login 2. Submit 3 queries 3. Attempt 4th	System blocks request with limit message
8	Submit 365-day forecast query (Enterprise)	1. Login 2. Fill 365-day query 3. Submit	Complete data with probabilistic overlays shown
9	Configure extreme event alert for heatwave	1. Login 2. Set >80% heatwave alert 3. Save	Alert is saved and will trigger if condition is met
10	Save and rerun previous query	1. Login 2. Save query 3. Go to history 4. Rerun	Query executes and displays updated forecast
11	Export forecast in Excel	1. Login 2. Run query 3. Click Export 4. Choose Excel	Data downloaded as formatted Excel file
12	Access dashboard as guest	1. Open site URL 2. Attempt dashboard access	Redirects to login with message
13	Subscribe to Professional Tier	1. Login 2. Go to Billing 3. Choose Tier 4. Submit payment	Payment processed and access upgraded

14	Receive email alert for drought	1. Login 2. Set drought alert 3. Wait for condition	User receives email alert
15	Search location using map interface	1. Open query 2. Use map to select location	Location populates from map selection
16	View probabilistic confidence graph	1. Run query with probabilistic option 2. View graph	Confidence intervals and overlays shown clearly
17	Compare two forecast models	1. Login 2. Choose Compare 3. Select providers	Forecasts shown side-by-side with highlights
18	Switch between light and dark mode	1. Login 2. Go to Settings 3. Toggle mode	UI theme updates and persists on next login
19	Download historical forecast report	1. Login 2. Go to History 3. Select past query 4. Click Download	Historical data downloaded in CSV or PDF